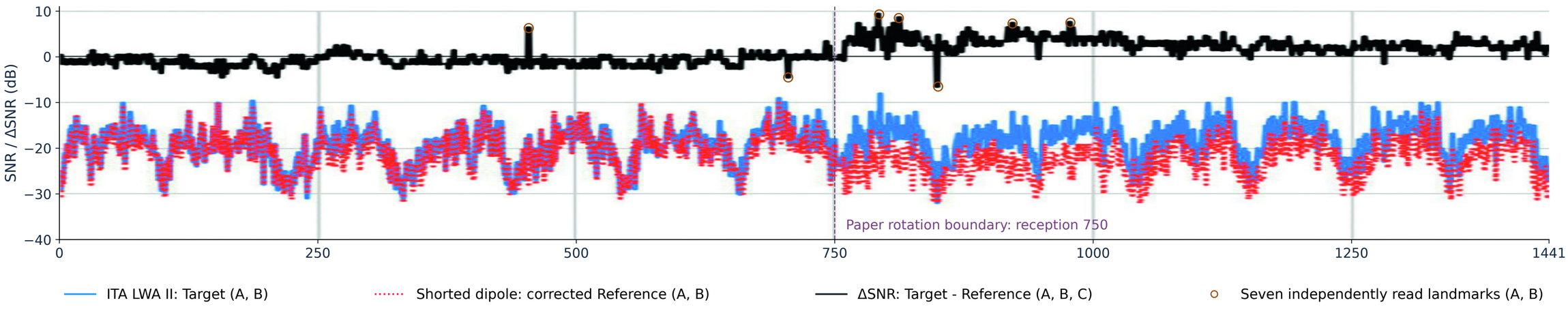
WSPRadar.org reconstruction & comparison

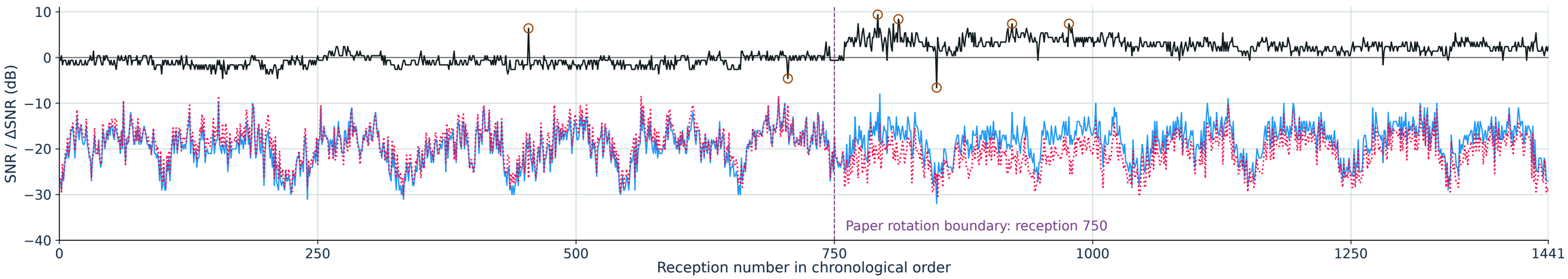
Antenna rotation: from individual receptions to 12-hour evidence

Referenced publication: J. Vanhamel, W. Machiels and H. Lamy (2022). *Using the WSPR Mode for Antenna Performance Evaluation and Propagation Assessment on the 160-m Band.*
Source figure: **Figure 6, p. 6** · International Journal of Antennas and Propagation (2022), Article 4809313
Demo: RX Benchmark | 160 m | 1-15 May 2021 | M7AEO (IO82) to Target ON4AWM0 (JO20OT) / Reference ON4AWM1 (JO20) | 1 441 spot pairs | Reference correction +1.6 dB

Panel A - Original image from publication

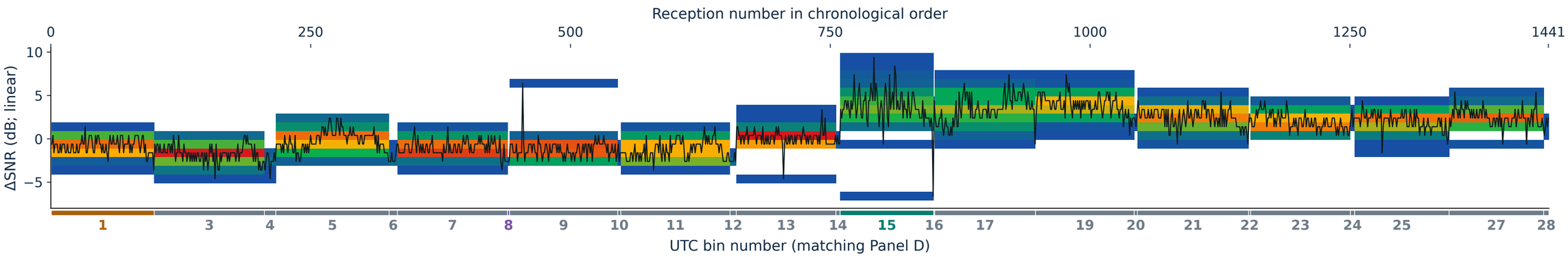


Panel B - Reconstruction: every paired reception



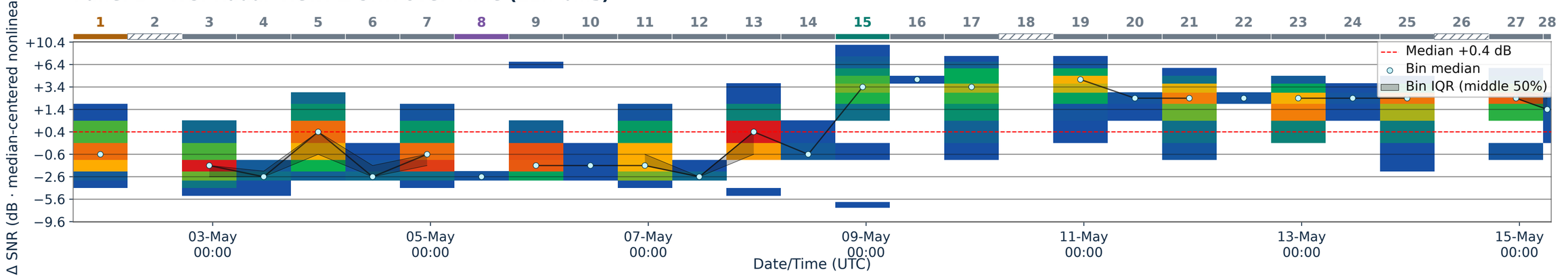
Panels A and B use identical linear axes and reported SNR for the antenna traces. No fitted shift, amplitude scaling or time warp.

Panel C - Linking the black ΔSNR curve to the 12-hour density bins



Each column is one UTC bin: Panel C counts receptions; Panel D shows elapsed time. Cells are 1 dB high; their centres and edges shift by the negative Reference correction.

Panel D - WSPRadar view: Δ SNR over Time (12h bins)



Matching bin numbers link Panels C and D. Empty bins (hatched in Panel D) have no width in Panel C. Final bin: 1 h 45 min.
Bins start at 17:15 / 05:15 UTC. Panel D retains its nonlinear dB axis, medians, IQR and tail cells. Reception 750 does not date the rotation.